

Problem Book

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Chapter 1

Integration Problems

1.1 Problem 1

$$\int \cos^n x \, dx = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

Proof:

To prove the reduction formula for $\int \cos^n x \, dx$, we use integration by parts. Consider the integral:

$$I_n = \int \cos^n x \, dx.$$

Rewrite $\cos^n x$ as $\cos^{n-1} x \cdot \cos x$. Let:

$$u = \cos^{n-1} x, \quad dv = \cos x \, dx.$$

Then,

$$\begin{aligned} du &= (n-1) \cos^{n-2} x \cdot (-\sin x) \, dx = -(n-1) \cos^{n-2} x \sin x \, dx, \\ v &= \sin x. \end{aligned}$$

Using integration by parts, $\int u \, dv = uv - \int v \, du$, we get:

$$I_n = \cos^{n-1} x \cdot \sin x - \int \sin x \cdot [-(n-1) \cos^{n-2} x \sin x] \, dx.$$

Simplify the expression:

$$I_n = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \sin^2 x \, dx.$$

Using the identity $\sin^2 x = 1 - \cos^2 x$, we have:

$$I_n = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x (1 - \cos^2 x) \, dx.$$

Split the integral:

$$I_n = \cos^{n-1} x \sin x + (n-1) \left(\int \cos^{n-2} x \, dx - \int \cos^n x \, dx \right).$$

Notice that $\int \cos^n x \, dx = I_n$, so:

$$I_n = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \, dx - (n-1) I_n.$$

Rearrange to solve for I_n :

$$I_n + (n-1) I_n = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \, dx,$$

$$nI_n = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x dx.$$

Finally, divide by n :

$$I_n = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x dx.$$

Thus, the reduction formula is:

$$\boxed{\int \cos^n x dx = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} \int \cos^{n-2} x dx.}$$